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1 Introduction

Family size is an important demographic variable influencing the economy, public policy decisions and the society. Understanding the determinants of family size can help the economist and policymakers addressing to make better resource allocation and population sustainability. This study investigates the influence of literacy and age at marriage on family size, using data from the World Fertility Survey conducted in Portugal between 1979 and 1980. These factors were selected due to their significant associations with birth rate, as highlighted in previous research.

Existing literature emphasizes the importance of socio-economic and cultural factors in shaping family size. Ahmed and Roy (2023) identified that variables such as family income, educational status, and contraceptive use significantly impact household size in Bangladesh. Their findings emphasize the role of education and socio-economic conditions, aligning with this study's focus on literacy as a determinant of family size. Vignoli et al. (2021) expanded this understanding by analyzing how educational pairings influence fertility transitions in European countries, showing that couples with higher educational attainment often have smaller families. Additionally, Bumpass and Mburugu (1977) highlighted the effects of age at marriage on family size, illustrating that marrying at very young or older ages significantly alters fertility outcomes.

This study examines how literacy and age at marriage jointly affect family size in Portugal, aiming to build on and expand existing research in the field. Specifically, the research objective is to quantify the impact of these variables on family size using data from the World Fertility Survey conducted in Portugal between 1979 and 1980. The peer reviewed studies in previous paragraph highlight the importance of literacy and age at marriage in shaping family size separately. By explicitly connecting these insights, this study seeks to provide a detailed understanding of how literacy and age at marriage, influence family size in Portugal.

To achieve this objective, the study uses a Poisson generalized linear model (GLM), which is a appropriated method for analyzing count data like family size. The Poisson GLM enables the modeling of relationships between predictor variables and a discrete, non-negative dependent variable. By using this method, the study aims to quantify the effects of these predictors on family size, bridging gaps identified in previous research. The application of the Poisson GLM

also strengthens the validity of the findings and let the results contribute meaningfully to the study.

2 Methodology

Given that our research objective is to analyze the factors influencing the number of children born, we initially considered Poisson regression as our modeling approach, as it is well-suited for count data. Since individuals have varying lengths of exposure time, we incorporated $\log(\text{Months married})$ as an offset, effectively transforming the model into a rate-based framework, where the response variable represents the average number of children born per month.

Reflecting on our research question: “How do literacy and age of marriage affect family size?” Given this focus, we determined that the variable “region” was not relevant to our study. Therefore, we excluded it from our model. Our initial model included all variables except for “region”, ensuring that our analysis remained centered on the key factors influencing family size.

We began by validating the assumptions of Poisson regression to ensure the appropriateness of our model. First, we confirmed that the log mean of the response variable should have a linear relationship with the predictor variables. Next, we verified the independence of observations. We also tested for equidispersion. If overdispersion was detected, we considered alternative models such as negative binomial regression to account for dispersion effects. Finally, we confirmed that our response variable is a non-negative integer counts variable.

We then assessed the correlation among numerical variables to identify potential multicollinearity. For pairs of variables with a strong correlation (correlation coefficient > 0.5), we retained only one representative variable to prevent redundancy and improve model stability. This refined model, incorporating only the selected predictors, is referred to as the parent model.

To refine our model, we systematically assessed the significance of each predictor variable. We generated a series of nested models, where each new model was derived by removing a single predictor from the parent model. To evaluate the impact of each predictor, we conducted likelihood ratio tests (LRTs), comparing each reduced model to the parent model. If the LRT yielded a p-value greater than 0.05, indicating that the removal of the predictor did not significantly reduce the model’s explanatory power, we considered the variable to be non-essential and excluded it from the final model.

3 Result

The following is our final model:

$$\begin{aligned}\log(E[\text{Children}_i]) = & \beta_0 + \beta_1 M_{0to15}(i) + \beta_2 M_{15to18}(i) + \\ & \beta_3 M_{18to20}(i) + \beta_4 M_{20to22}(i) + \beta_5 M_{25to30}(i) + \\ & \beta_6 M_{30toInf}(i) + \beta_7 \text{Pregnancies}_i + \beta_8 \text{Literacy}_i + \\ & \log(\text{Months}_i) + \epsilon_i\end{aligned}$$

Where :

Children_i : Number of children born to the i th individual.

$M_{atob}(i)$: Binary indicator variable, where $M_{atob}(i) = 1$
if the i th individual was married between ages a and b .

Pregnancies_i : Total number of pregnancies experienced by the i th individual.

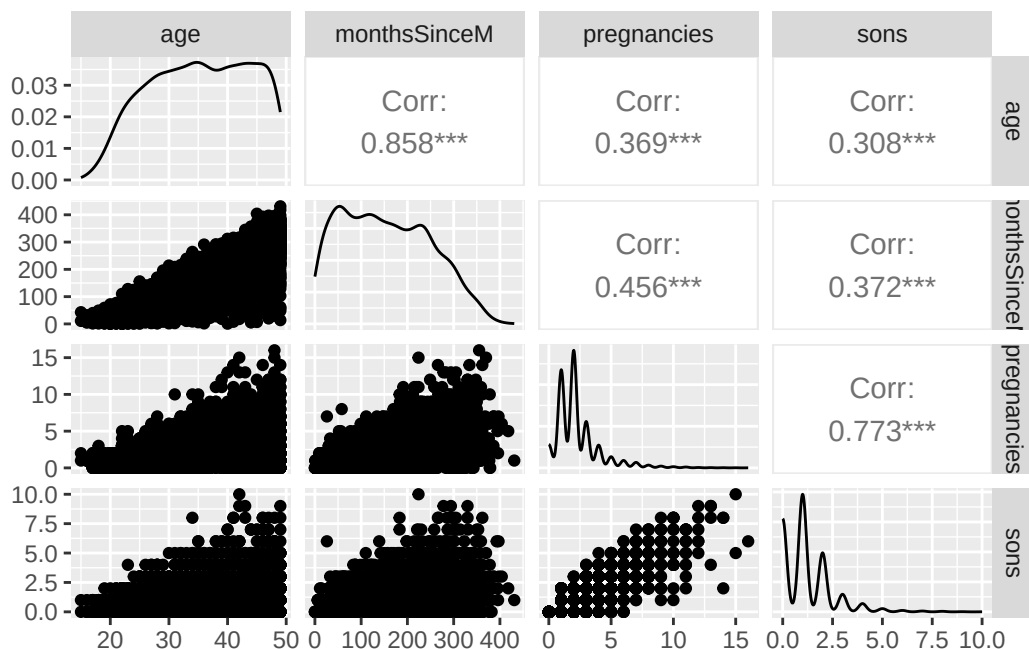
Literacy_i : Binary variable, where $\text{Literacy}_i = 1$
indicates that the i th individual is illiterate.

Months_i : Total number of months since marriage.

$\beta_0, \beta_1, \dots, \beta_8$: Regression coefficients estimated in the model.

ϵ_i : Error term accounting for random deviations.

```
subset_data <- subset(portugal, select = c(age, monthsSinceM, pregnancies, sons))
ggpairs(subset_data)
```



```
summary(portugal)
```

age	ageMarried	monthsSinceM	pregnancies
Min. :15.00	22to25 :1468	Min. : 0.0	Min. : 0.000
1st Qu.:28.75	20to22 :1126	1st Qu.: 73.0	1st Qu.: 1.000
Median :36.00	25to30 : 923	Median :148.0	Median : 2.000
Mean :35.30	18to20 : 910	Mean :155.2	Mean : 2.282
3rd Qu.:43.00	15to18 : 452	3rd Qu.:230.0	3rd Qu.: 3.000
Max. :49.00	30toInf: 217	Max. :431.0	Max. :16.000
	(Other): 52		
children	sons	region	literacy
Min. : 0.00	Min. : 0.000	lt10k :3502	yes:4567
1st Qu.: 1.00	1st Qu.: 0.000	lisbon: 470	no : 581
Median : 2.00	Median : 1.000	porto : 160	
Mean : 2.26	Mean : 1.187	20k+ : 583	
3rd Qu.: 3.00	3rd Qu.: 2.000	10-20k: 433	
Max. :17.00	Max. :10.000		

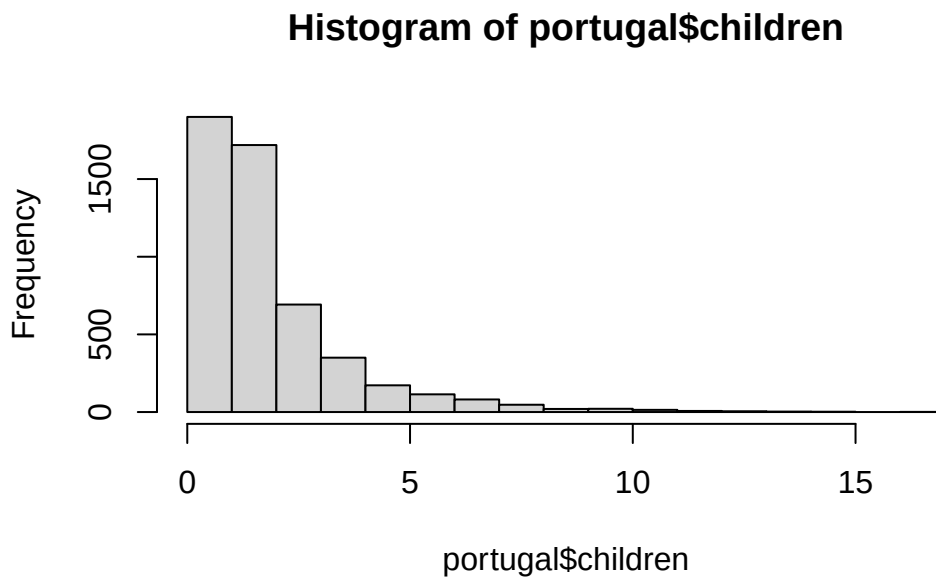
```
# Check for overdispersion.
mean(portugal$children)
```

```
[1] 2.26049
```

```
sd(portugal$children)^2
```

```
[1] 3.463704
```

```
# Check for zero inflation.  
hist(portugal$children)
```



```
model_parant <- glm(children ~ ageMarried + pregnancies + literacy ,  
  family = poisson(link = "log"),  
  data = portugal,  
  offset = log(monthsSinceM + 0.00000000001))  
  
model_noageMarried <- glm(children ~ pregnancies + literacy ,  
  family = poisson(link = "log"),  
  data = portugal,  
  offset = log(monthsSinceM + 0.00000000001))  
  
model_nopregnancies <- glm(children ~ ageMarried + literacy ,  
  family = poisson(link = "log"),
```

```

        data = portugal,
        offset = log(monthsSinceM + 0.000000000001))

model_noliteracy <- glm(children ~ ageMarried + pregnancies,
        family = poisson(link = "log"),
        data = portugal,
        offset = log(monthsSinceM + 0.000000000001))

```

```

# Define all models to be tested
models <- list(
  "model_noageMarried" = model_noageMarried,
  "model_nopregnancies" = model_nopregnancies,
  "model_noliteracy" = model_noliteracy
)

# Create an empty data frame to store p-values
p_values <- data.frame(Model = character(), P_Value = character(), stringsAsFactors = FALSE)

# Perform the likelihood ratio test for each model against model_parant
for (name in names(models)) {
  test_result <- lrtest(model_parant, models[[name]]) # Perform likelihood ratio test

  p_value <- as.numeric(test_result$"Pr(>Chisq)"[2]) # Extract p-value as a numeric value

  # Format the p-value to scientific notation with 3 decimal places before 'e'
  formatted_p_value <- formatC(p_value, format = "e", digits = 3)

  # Append the result to the data frame
  p_values <- rbind(p_values, data.frame(Model = name, P_Value = formatted_p_value))
}

# Print the final data frame containing model names and corresponding p-values
print(p_values)

```

```

      Model    P_Value
1 model_noageMarried 1.984e-05
2 model_nopregnancies 0.000e+00
3   model_noliteracy 3.758e-15

```

```
summary(model_parant)
```

Call:

```
glm(formula = children ~ ageMarried + pregnancies + literacy,  
     family = poisson(link = "log"), data = portugal, offset = log(monthsSinceM +  
                           1e-12))
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.722750	0.020743	-227.676	< 2e-16 ***
ageMarried0to15	-0.054594	0.079488	-0.687	0.492201
ageMarried15to18	-0.077782	0.034227	-2.273	0.023053 *
ageMarried18to20	-0.063742	0.027951	-2.280	0.022579 *
ageMarried20to22	-0.023768	0.026471	-0.898	0.369245
ageMarried25to30	0.051445	0.028722	1.791	0.073274 .
ageMarried30toInf	0.195177	0.058864	3.316	0.000914 ***
pregnancies	0.167649	0.003425	48.947	< 2e-16 ***
literacyno	-0.195656	0.025282	-7.739	1e-14 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 5654.3 on 5147 degrees of freedom
Residual deviance: 3607.1 on 5139 degrees of freedom
AIC: 16056

Number of Fisher Scoring iterations: 5